Form Number : ____________

ALLEN CAREER INSTITUTE

PRELIMINARY EXAM : 2019-20

Paper Set: SET-I (HT) | **SUBJECT: Maths I (Algebra)** | **Max. Marks: 40**

SSC Board - Sample Paper - 1 | **Duration: 2 Hrs.**

Entire Syllabus

Q.1(A) Choose the correct alternative **4M**

i) 210 is the __________ term of the A.P.21,42,63,84......

a) 10^{th} **b)** 11^{th} **c)** 12^{th} **d)** 13^{th}

ii) To find the cost of one share at the time of buying, the amount of Brokerage and GST is to be _________ MV of share.

a) Added to **b)** subtracted form

c) Multiplied with **d)** divided by

iii) The formula to find mean from a grouped frequency table is $X = A + \frac{\sum f_i u_i}{\sum f_i} \times g$. in the formula

$u_i =$ ______.

a) $\frac{x_i + A}{g}$ **b)** $(x_i - A)$ **c)** $\frac{x_i - A}{g}$ **d)** $\frac{A - x_i}{g}$

iv) To draw graph of $4x - 5y = 19$, find y when x = 1.

a) 4 **b)** 3 **c)** 2 **d)** - 3

Q.1 (B) Solve the following questions. **4M**

i) Find the value of discriminant of the quadratic equation $5m^2 - m = 0$.

ii) On certain article if rate of CGST is 9%, then what is the rate of SGST? And what is the rate of GST?

iii) Different expenditures incurred on the construction of a building were shown by a pie diagram. The expenditure of Rs 45,000 on cement was shown by a sector of central angle of 75^0. What was the total expenditure of the construction?

iv) There are 15 tickets bearing the numbers from 1 to 15 in a bag and one ticket is drawn from this bag at random. Write the sample space (S) and n (S).

Q.2 A Complete the following activities. (Any two) **4M**

i) Fill up the boxes and find out the number of terms in the A.P.2,4,6….,148.

Here, a = 2, d $\boxed{}$, $t_n = 148$

$t_n = a + (n-1)d$

$\therefore 148 = \boxed{}$

$\therefore 146 = 2n - \boxed{}$

$\therefore n = \boxed{}$

ii Complete the following activity to form a quadratic equation. Activity:

I am a quadratic equation

↓

My standard form is $\boxed{}$

↓

My roots are 3 and 4.

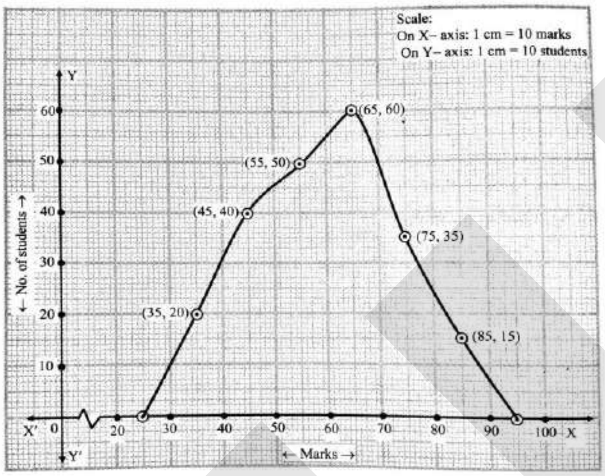
↓

Sum of my roots = $\boxed{}$
And product of my roots = $\boxed{}$

↓

My equation is $\boxed{}$

iii)



Observe the above frequency polygon and fill in the following boxes.

i. The class $\boxed{}$ has the maximum number of students.

ii. The classes 20-30 and $\boxed{}$ have frequency zero.

iii. The class mark of the class having 50 students is $\boxed{}$

iv. There are $\boxed{}$ students in the class 80 – 90.

Q.2 B Solve the following questions (Any four) **8M**

i. The taxable value of a wrist watch belt is Rs 586. Rate of GST is 18%. Then what is price of the belt for the customer?

ii. The annual investments of a family are shown in the given pie diagram. Answer the following questions based on it.
a. If the investment in shares is Rs 2000, find the total investment.
b. How much amount is deposited in bank?

iii Find the A.P whose first term is -19 and common difference is -4.

iv. Smt. Deshpande purchased of FV Rs 5 at a premium of Rs 20. How many shares will she get for Rs 20,000?

v. Obtain a quadratic equation whose roots are -3 and -7.

Q.3 A Complete the following activities (Any one) **3M**

i To solve given equations, fill the below boxes suitably.

$$\frac{5}{x-1}+\frac{1}{y-2}=2;\ \frac{6}{x-1}-\frac{3}{y-2}=1$$

Replacing $\left(\frac{1}{x-1}\right) by\, m,\ \left(\frac{1}{y-2}\right) by\, n$

New equations

$\boxed{}$ 6m-3m=1

On solving

m = $\boxed{}$, n = $\boxed{}$

Replacing m, n by their original values.

$\frac{1}{x-1}=\frac{1}{3}$ $\boxed{}$

On solving

X = $\boxed{}$, y = $\boxed{}$

ii A frequency distribution table for the production of oranges of some farm owners is given below. To find the mean production of oranges by 'assumed mean' method, fill in the blanks.

Class Production (Thousand rupees)	25- 30	30- 35	35 – 40	40 – 45	45 – 50
No. of farm owners	20	25	15	10	10

Class Production (Thousand rupees)	Class mark xi	$d_i = xi - A = xi - 37.5$	Frequency (No. of farm owners)	Frequency X Deviation *fidi*
25 – 30	$\boxed{}$	- 10	20	- 200
30 – 35	32.5	-5	25	- 125
35 – 40	37.5	0	15	0
40 – 45	42.5	5	10	50
45 – 50	47.5	$\boxed{}$	10	100
Total	-	-	$\sum fi = 80$	$\sum fidi = \boxed{}$

$\bar{d} = \boxed{}$

Mean = $\bar{X} = A + \boxed{}$ .... [Formula]

= 35.31 thousand

= Rs $\boxed{}$

Q3.B Solve the following questions (Any two) **6M**

Joseph purchased following shares, Find his total investment.
Company A : 200 shares , FV = Rs 2, Premium = Rs 18.
Company B : 45 shares, MV = Rs 500
Company C : 1 share, MV = Rs 10, 540

ii. If the 9th term of an A.P. is zero, then show that the 29th term is twice the 19th term.

iii. Sum of the roots of a quadratic equation is double their product. Find k if equation is $x^{-2} - 4kx + k + 3 = 0.$

iv In a game of chance, a spinning arrow come to rest at one of the numbers 1,2,3,4,5,6,7,8. All these are equally likely outcomes. Find the probability that it will rest at

a. an odd number. **b.** a number greater than 2. **c.** a number less than 9.

Q.4 Solve the following questions. (Any two) **8M**

i. Sum of areas of two squares is 244 cm² and the difference between their perimeter is 8 cm. Find the ratio of their diagonals.

ii. Draw the graphs representing the equations $4x + 3y = 24 and 3y = 4x + 24$ on the same graph paper. Find the area of the triangle formed by these lines and the X-axis.

iii The median of the following data is 50. Find the values of P and Q, if the sum of all the frequency is 90.

Marks	**Frequency**
20 – 30	P
30 – 40	15
40 – 50	25
50 – 60	20
60 – 70	Q
70 – 80	8
80 – 90	10

Q.5 Solve the following questions. (Any one) **3M**

i The following frequency distribution table shown the distances travelled by some rickshaws in a day. Observe the table and answer the following questions:

Class (Daily distance travelled in km)	**Continuous Classes**	**Frequency (Number or rickshaws)**	**Cumulative Frequency less than type**
60 – 64	59.5 – 64.5	10	10
65 – 69	64.5 – 69.5	34	10+34 = 44
70 – 74	69.5 – 74.5	58	44+58 = 102
75 – 79	74.5 – 79.5	82	102 + 82= 184
80 – 84	79.5 – 84.5	10	184 + 10 = 194
85 - 89	84.5 – 89.5	6	194 + 6 = 200

a. Which is the modal Class? Why?
b. Which is the median class and why?
c. Write the cumulative frequency (C.F.) of the class preceding the median class.
d. What is the class interval (h) to calculate median?

ii. There are three dealers A, B and C in Maharashtra. Suppose, the trade of each of them in September 2018 was as shown in the following table. The rate of GST on each transaction was 5%. Read the table and answer the questions below it.

Dealer	GST collected on the sale	GST paid at the time of purchase	ITC	Tax paid to the Govt.	Tax- balance with the Govt.
A	Rs . 5,000	Rs. 6,000	Rs. 5,000	Rs 0	Rs 1,000
B	Rs. 5,000	Rs, 4,000	Rs, 4,000	Rs 1,000	Rs 0
C	Rs. 5000	Rs, 5,000	Rs, 5,000	Rs 0	Rs 0

a. How much amount did the dealer A get by sale?
b. For how much amount did the dealer B buy the articles?
c. How much is the balance of CGST and SGST left with the government that was paid by A?